CONFIDENTIAL — INVESTOR-2026

Cardiocerebral Coupling and Linguistic Surveillance: A Neural Oscillation Signature Theory Framework for Early Detection of Cardiovascular Pathology

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The author is the founder and president of Beverly Index LLC, a company developing commercial applications based on the theoretical framework presented in this paper. This potential conflict of interest is disclosed in accordance with APA ethical standards. The theoretical framework and proposed study design presented herein are offered for scientific scrutiny and collaborative development by the research community.

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Abstract

Cardiovascular disease remains the leading cause of death globally, yet the majority of first acute coronary events occur in individuals without prior cardiovascular diagnosis. Current risk stratification methods—lipid panels, coronary calcium scoring, stress testing—assess structural and biochemical risk factors but do not directly measure whether progressive cardiac pathology is producing functional consequences in end-organ systems. This paper proposes a novel detection framework based on Neural Oscillation Signature Theory (NOST; Beverly,

CONFIDENTIAL — INVESTOR-2026

CONFIDENTIAL — INVESTOR-2026

2026b), exploiting the bidirectional coupling between cardiac function and cerebral neural oscillatory activity. Five pathophysiological pathways are identified through which progressive cardiovascular disease produces measurable changes in cerebral oscillatory dynamics: (1) reduced cardiac output and cerebral hypoperfusion in subclinical heart failure, (2) subclinical microembolic burden in atrial fibrillation and valvular disease, (3) cardiac autonomic dysregulation reflected in altered central autonomic network oscillatory activity, (4) systemic inflammation and endothelial dysfunction affecting cerebrovascular reactivity, and (5) shared atherosclerotic burden producing simultaneous coronary and cerebral vascular compromise. Because NOST proposes that neural oscillatory states leave detectable signatures in linguistic output, these cardiac-origin cerebral changes should manifest as measurable shifts in the statistical properties of language. This paper articulates the detection chain for each pathway, specifies the predicted linguistic signatures, and proposes a prospective longitudinal validation study in a cardiac risk cohort. If confirmed, the framework would establish language-based monitoring as a functional early warning system for cardiovascular events—a scalable, non-invasive complement to existing structural and biochemical risk assessment.
Keywords: cardiovascular disease, cardiocerebral coupling, neural oscillations, linguistic markers, NOST, NOSS, early detection, heart failure, atrial fibrillation, autonomic dysregulation, cardiac screening

Cardiocerebral Coupling and Linguistic Surveillance:

A Framework for Early Detection of Cardiovascular Pathology

Cardiovascular disease (CVD) is the leading cause of death worldwide, responsible for approximately 17.9 million deaths annually (World Health Organization, 2021). In the United States alone, someone experiences a myocardial infarction every 40 seconds, and coronary heart

CONFIDENTIAL — INVESTOR-2026

disease accounts for approximately one in every seven deaths (Virani et al., 2021). Despite decades of progress in risk factor identification and management, the clinical reality remains stark: the majority of first acute coronary events—including myocardial infarction and sudden cardiac death—occur in individuals who were not previously identified as high-risk by conventional screening methods (Khot et al., 2003; Greenland et al., 2010).

This detection gap reflects a fundamental limitation of current cardiovascular risk assessment: existing methods primarily measure structural and biochemical risk factors (lipid levels, coronary calcium, blood pressure, blood glucose) rather than the functional consequences of progressive cardiac pathology on end-organ systems. A patient may carry a significant coronary atherosclerotic burden, progressive myocardial dysfunction, or intermittent arrhythmia for months or years before these conditions produce symptoms severe enough to trigger clinical evaluation. During this prodromal interval, the pathology is advancing—and the brain, as one of the most metabolically demanding and perfusion-sensitive organs, is among the first to register its consequences.

This paper proposes that the cerebral consequences of progressive cardiovascular disease can be detected through the analysis of linguistic output, using the framework provided by Neural Oscillation Signature Theory (NOST; Beverly, 2026b). NOST establishes that neural oscillatory states systematically shape language production, such that the statistical properties of spoken and written text carry extractable information about the oscillatory configuration that generated them. If cardiac pathology alters cerebral oscillatory activity—and the evidence reviewed below demonstrates that it does, through multiple well-documented pathways—then those alterations should produce detectable linguistic signature changes amenable to computational analysis.

CONFIDENTIAL — INVESTOR-2026

A companion paper (Beverly, 2026e) addresses the detection of subclinical cerebrovascular compromise through linguistic monitoring. The present paper addresses a distinct but complementary clinical domain: the detection of primary cardiac pathology through its secondary cerebral effects. The two frameworks share the final links of their detection chains (oscillatory disruption → linguistic signature change) but differ in their entry points and target pathology, and they address different clinical populations, different medical specialties, and different intervention pathways.

The Cardiocerebral Coupling Principle

The heart and brain are not independent organs operating in physiological isolation. They are bidirectionally coupled through multiple mechanisms—hemodynamic, autonomic, inflammatory, and embolic—that ensure that changes in cardiac function produce measurable changes in cerebral function, and vice versa. This coupling is sufficiently robust that the field of neurocardiology has emerged to study the bidirectional interactions between cardiovascular and neurological disease (Samuels, 2007; Silvani et al., 2016). The present framework exploits this coupling in a specific direction: using cerebral functional consequences, detected through linguistic output, as indicators of cardiac pathology.

The cardiocerebral coupling principle can be stated formally: any cardiac pathology that alters (a) the volume of blood delivered to the brain, (b) the quality of blood delivered to the brain (embolic content, oxygenation, inflammatory mediators), or (c) the autonomic regulatory signals between the heart and central nervous system will produce measurable changes in cerebral neural oscillatory activity. If NOST's central claim is correct—that oscillatory changes produce detectable linguistic signature changes—then cardiac pathology should be linguistically

CONFIDENTIAL — INVESTOR-2026

detectable to the extent that it produces cerebral consequences through any of these three channels.

Five Pathophysiological Pathways

This section identifies five distinct pathways through which progressive cardiovascular disease produces cerebral oscillatory changes detectable through NOST-based linguistic monitoring. Each pathway is supported by established empirical evidence linking the cardiac pathology to documented cerebral consequences.

Pathway 1: Reduced Cardiac Output and Cerebral Hypoperfusion

The brain receives approximately 15% of cardiac output and consumes approximately 20% of the body's oxygen supply. Neural tissue is exquisitely sensitive to reductions in perfusion: even modest decreases in cerebral blood flow produce measurable changes in neural function before structural damage occurs. Heart failure—whether with reduced ejection fraction (HFrEF) or preserved ejection fraction (HFpEF)—produces chronic reductions in cardiac output that directly compromise cerebral perfusion.

The cognitive consequences of heart failure are now well established. A meta-analysis by Cannon et al. (2011) documented that heart failure patients show significant deficits in attention, executive function, processing speed, and memory compared with age-matched controls. Critically, these cognitive deficits are present even in compensated heart failure—patients who are clinically stable and asymptomatic at the time of testing—suggesting that subclinical cerebral hypoperfusion produces functional consequences that precede clinical heart failure symptoms (Pressler et al., 2010).

The electroencephalographic consequences of cardiac-output-dependent cerebral hypoperfusion include: generalized slowing of background activity (increased delta and theta

CONFIDENTIAL — INVESTOR-2026

power), reduced alpha-band peak frequency, and diminished beta-band modulation during cognitive tasks (Vogels et al., 2007). These oscillatory changes overlap substantially with those produced by cerebrovascular disease (Beverly, 2026e), but the clinical context differs: in heart failure, the cerebral perfusion deficit is global and cardiac-output-dependent rather than focal and vessel-territory-dependent. This distinction predicts a diffuse, bilateral NOSS activation pattern (primarily S4 Neurocognitive) rather than the lateralized or focal pattern expected in cerebrovascular disease.

The linguistic consequences predicted by NOST for subclinical cardiac-output-dependent hypoperfusion include: reduced syntactic complexity reflecting compromised executive function, slowed processing reflected in increased pause frequency and reduced speech rate, diminished information density reflecting impaired working memory capacity, and reduced narrative coherence reflecting compromised attentional resources. These predictions are consistent with the neuropsychological profile documented in heart failure patients and are amenable to testing through computational linguistic analysis.

Pathway 2: Subclinical Microembolic Burden

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia, affecting approximately 33 million individuals worldwide (Chugh et al., 2014). AF is well established as a major risk factor for stroke through macroscopic thromboembolism, but the cerebral consequences of AF extend far beyond clinical stroke events. Transcranial Doppler studies have documented that AF patients experience subclinical cerebral microemboli at rates far exceeding clinical event rates—high-intensity transient signals indicative of microembolic events occur in a substantial proportion of AF patients, even those receiving anticoagulation therapy (Gaita et al., 2013).

CONFIDENTIAL — INVESTOR-2026

These subclinical microemboli produce cumulative cerebral damage detectable on neuroimaging as silent cerebral infarcts and white matter lesions. The cognitive consequences are clinically meaningful: a systematic review and meta-analysis by Kalantarian et al. (2013) found that AF is independently associated with cognitive impairment and dementia even in the absence of clinical stroke, with a relative risk of cognitive decline approximately 40% higher in AF patients than controls. The mechanism appears to be cumulative microembolic and hypoperfusion-mediated damage to white matter tracts and subcortical circuits supporting executive function and processing speed.

The oscillatory consequences of AF-related cerebral microembolic burden include: increased theta-band power reflecting subcortical white matter disruption, slowed dominant alpha frequency, and reduced coherence between frontal and parietal regions reflecting compromised long-range connectivity (Moretti et al., 2012). NOST predicts that these oscillatory changes should produce a linguistic signature characterized by executive-attentional impairment: reduced discourse planning, increased topic fragmentation, and diminished coherence maintenance across extended narrative—a pattern detectable through serial linguistic monitoring.

Valvular heart disease, particularly mechanical prosthetic valves and calcific aortic stenosis, represents an additional source of subclinical microembolic burden. Transcranial Doppler studies have documented microembolic signals in a substantial proportion of patients with mechanical valve prostheses (Daffertshofer et al., 2004). The cerebral consequences parallel those of AF-related microembolism and are predicted to produce similar linguistic signature changes.

Pathway 3: Cardiac Autonomic Dysregulation

CONFIDENTIAL — INVESTOR-2026

This pathway represents the most novel and potentially most powerful detection mechanism proposed in this paper. The heart and brain are bidirectionally coupled through the autonomic nervous system: parasympathetic innervation via the vagus nerve, sympathetic innervation via the stellate ganglia, and central regulation through the central autonomic network (CAN) encompassing the insular cortex, anterior cingulate cortex, amygdala, hypothalamus, and brainstem nuclei (Benarroch, 2014). Cardiac autonomic dysfunction—reflected in reduced heart rate variability (HRV), impaired baroreflex sensitivity, and sympathovagal imbalance—is among the strongest predictors of adverse cardiovascular outcomes, including sudden cardiac death, myocardial infarction, and heart failure progression (La Rovere et al., 2003; Thayer et al., 2012).

The critical insight for the present framework is that the central autonomic network is not a dedicated cardiac control circuit—it is integrated with the broader neural architecture supporting cognition, emotion, and language production. The insular cortex participates in interoceptive awareness and emotional processing. The anterior cingulate cortex is central to cognitive control and conflict monitoring. The amygdala modulates threat processing and emotional regulation. Dysfunction in these structures—whether caused by primary cardiac autonomic pathology or by the bidirectional consequences of cardiac disease on central autonomic regulation—directly affects the neural oscillatory patterns that NOST proposes to detect through linguistic analysis.

Thayer and Lane's (2009) neurovisceral integration model provides the theoretical bridge: vagally-mediated heart rate variability reflects the functional integrity of the prefrontal-subcortical inhibitory circuits that support cognitive flexibility, attentional control, and emotional regulation. Reduced HRV is associated not only with cardiovascular risk but with measurable deficits in executive function, working memory, and attentional control (Hansen et al., 2003).

CONFIDENTIAL — INVESTOR-2026

The oscillatory correlates are specific: vagal tone is positively associated with alpha-band power and alpha-band reactivity, while sympathetic dominance is associated with increased beta-band activity and reduced alpha modulation (Chang et al., 2013).

NOST predicts that cardiac autonomic dysregulation should produce a distinctive multi-sector NOSS activation pattern: primary S4 (Neurocognitive) elevation reflecting executive-attentional compromise mediated by prefrontal-subcortical dysfunction, with possible secondary S2 (Anxiety) elevation reflecting sympathetic dominance and altered threat-processing dynamics, and possible S1 (Mood) elevation reflecting the well-documented association between autonomic dysregulation and affective disturbance. This multi-sector pattern—executive-attentional impairment accompanied by affective-autonomic signatures—would be distinguishable from the pure neurocognitive pattern predicted for cardiac-output-dependent hypoperfusion (Pathway 1) and the executive-predominant pattern predicted for microembolic burden (Pathway 2).

The linguistic consequences predicted for cardiac autonomic dysregulation include: reduced cognitive flexibility reflected in decreased syntactic variety and increased reliance on habitual constructions, diminished attentional control reflected in increased susceptibility to tangential content and reduced relevance maintenance, increased emotional reactivity reflected in elevated negative valence vocabulary and increased absolutist language markers, and altered prosodic dynamics—particularly reduced pitch variability and altered speech rhythm—reflecting diminished vagal modulation of the laryngeal and respiratory musculature involved in speech production. This last prediction is especially significant because prosodic changes may provide the earliest and most sensitive indicator of autonomic dysregulation, detectable before cognitive-linguistic changes reach statistical significance.

CONFIDENTIAL — INVESTOR-2026

Pathway 4: Systemic Inflammation and Endothelial Dysfunction

Atherosclerosis is fundamentally an inflammatory disease (Libby et al., 2002). The progression from endothelial dysfunction to fatty streak formation to vulnerable plaque development is driven by inflammatory mediators including C-reactive protein (CRP), interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF-α), and matrix metalloproteinases. These inflammatory processes are systemic: they affect the cerebral vasculature simultaneously with the coronary vasculature, and they produce direct neuroinflammatory consequences that alter neural function independently of perfusion changes.

The cerebral consequences of chronic systemic inflammation are documented across multiple domains. Elevated CRP and IL-6 are associated with reduced processing speed and executive function in otherwise healthy adults (Marsland et al., 2015). Chronic inflammation is associated with EEG changes including increased slow-wave activity and reduced alpha-band power (Lasselin et al., 2020). Peripheral inflammatory challenge (e.g., vaccination-induced inflammation) produces acute cognitive changes and measurable shifts in neural oscillatory dynamics (Harrison et al., 2014)—demonstrating that the immune-to-brain signaling pathway is sufficiently fast and robust to produce detectable cognitive effects within hours of peripheral inflammatory activation.

NOST predicts that the chronic low-grade inflammation characteristic of progressive atherosclerotic disease should produce gradual linguistic changes reflecting its cognitive consequences: reduced processing efficiency, diminished executive capacity, and potential affective changes reflecting the well-documented association between chronic inflammation and depressive symptomatology (Miller & Raison, 2016). The linguistic signature would overlap with but be distinguishable from the hypoperfusion signature (Pathway 1) because inflammatory-

CONFIDENTIAL — INVESTOR-2026

mediated cognitive changes are less dependent on global perfusion and more closely associated with neuroinflammatory markers of executive and motivational impairment.

Pathway 5: Shared Atherosclerotic Burden

Atherosclerosis is a systemic vascular disease that affects multiple vascular beds simultaneously. Patients with significant coronary artery disease have markedly elevated rates of concurrent cerebrovascular disease, peripheral arterial disease, and renal artery stenosis (Bhatt et al., 2006). The corollary is that cerebral atherosclerotic changes—detectable through their cognitive and oscillatory consequences—serve as an indicator of systemic atherosclerotic burden that includes coronary involvement.

This pathway does not require a direct causal link from cardiac pathology to cerebral dysfunction. Rather, it exploits the epidemiological reality that the same pathological process producing coronary atherosclerosis is simultaneously producing cerebral atherosclerosis, with the latter detectable through NOST-based linguistic monitoring while the former remains clinically silent. Cerebral atherosclerosis produces the same subclinical small vessel disease and associated oscillatory changes described in the companion paper on cerebrovascular detection (Beverly, 2026e). The clinical implication is that a linguistic trajectory change consistent with subclinical cerebrovascular compromise should trigger not only cerebrovascular evaluation but comprehensive cardiovascular risk assessment—because the cerebral findings may be the first clinically detectable signal of a systemic process that includes coronary involvement.

Population-level data support this approach. The Rotterdam Study demonstrated that cerebral small vessel disease markers on MRI predict cardiovascular events (including myocardial infarction) independently of traditional cardiovascular risk factors (Defined by Debette et al., 2010; Ikram et al., 2012). If NOST can detect the cerebral manifestations of

CONFIDENTIAL — INVESTOR-2026

subclinical small vessel disease—as proposed in the companion paper—it thereby provides indirect detection of the shared atherosclerotic process that underlies coronary risk.

The Integrated Cardiac Linguistic Signature

The five pathways described above do not operate in isolation. Progressive cardiovascular disease typically involves multiple pathways simultaneously: a patient with heart failure may also have atrial fibrillation, autonomic dysregulation, systemic inflammation, and generalized atherosclerosis. The linguistic signature of progressive cardiac disease is therefore predicted to be a composite reflecting the relative contributions of each pathway in the individual patient.

The integrated cardiac linguistic signature is predicted to include the following core features, arranged by expected sensitivity and temporal precedence:

Earliest detectable changes: prosodic alterations reflecting autonomic dysregulation (reduced pitch variability, altered speech rhythm, changes in respiratory patterning during speech production). These changes reflect Pathway 3 and are predicted to precede cognitive-linguistic changes because autonomic dysregulation occurs early in cardiac disease progression and has direct effects on the motor systems supporting speech.

Early-intermediate changes: reduced processing efficiency reflected in increased pause frequency, slowed speech rate, reduced information density, and increased filler and hesitation marker usage. These changes reflect the combined effects of Pathways 1 (reduced cardiac output), 2 (microembolic burden), and 4 (systemic inflammation) on the processing speed and attentional resources that support fluent language production.

Intermediate changes: executive-linguistic degradation reflected in reduced syntactic complexity, diminished discourse planning, impaired coherence maintenance, and increased

CONFIDENTIAL — INVESTOR-2026

topic fragmentation. These changes reflect the cumulative impact of all five pathways on the frontal-subcortical circuits supporting executive function.

Later changes: affective-linguistic shifts reflected in increased negative valence vocabulary, altered emotional regulation markers, and potential increases in absolutist language—reflecting the well-documented comorbidity between progressive cardiac disease and depressive symptomatology, mediated by both inflammatory (Pathway 4) and autonomic (Pathway 3) mechanisms.

The temporal sequencing of these predicted changes is itself diagnostically informative. A pattern in which prosodic changes precede executive changes, which in turn precede affective changes, would be consistent with a cardiac-origin cognitive decline trajectory. A pattern in which affective changes appear first would suggest primary psychiatric pathology. A pattern in which executive changes appear without prosodic antecedents would suggest cerebrovascular rather than cardiac etiology. These differential trajectory signatures, if confirmed, would provide etiological information beyond simple detection.

Predicted NOSS Activation Pattern for Cardiac Pathology

The Neural Oscillatory Spectroscopy System (NOSS; Beverly, 2026c) provides a 12-sector classification framework for organizing cognitive state information extracted through NOST-based linguistic analysis. The predicted NOSS activation pattern for progressive cardiac pathology involves primary and secondary sector activations that are distinguishable from other clinical presentations.

Primary activation: S4 (Neurocognitive)—reflecting the executive, attentional, and processing speed deficits produced by all five cardiac pathways. The S4 activation is predicted to

CONFIDENTIAL — INVESTOR-2026

show an executive-predominant, processing-speed-sensitive profile distinguishable from the memory-predominant profile associated with Alzheimer's-type neurodegeneration (S8).

Secondary activation: S2 (Anxiety)—reflecting sympathetic dominance and altered threat-processing associated with cardiac autonomic dysregulation (Pathway 3). This secondary activation is predicted to manifest as physiological hyperarousal markers rather than the content-specific threat language characteristic of primary anxiety disorders.

Tertiary activation: S1 (Mood)—reflecting the affective consequences of chronic cardiac disease, mediated by inflammatory and autonomic pathways. This activation is predicted to appear later in the disease trajectory and to correlate with inflammatory markers.

The distinguishing feature of the cardiac NOSS pattern is the presence of prosodic changes—particularly reduced pitch variability and altered speech rhythm—in the context of executive-predominant S4 activation. Pure cerebrovascular disease (without significant cardiac involvement) is not predicted to produce the prosodic-autonomic signature because it does not directly involve the autonomic regulatory pathways that modulate laryngeal and respiratory function during speech production. This prosodic discriminator, if confirmed, would provide the basis for distinguishing cardiac-origin from cerebrovascular-origin cognitive-linguistic decline.

Proposed Validation Study

Study Design

The proposed study is a prospective longitudinal cohort design with a 48-month follow-up period. The extended follow-up (relative to the 36-month design proposed for cerebrovascular detection) reflects the typically longer prodromal period for cardiac events compared with cerebrovascular events, and the need to accumulate sufficient endpoint events for statistical analysis.

CONFIDENTIAL — INVESTOR-2026

Participants

Four cohorts would be recruited: (1) a cardiac risk cohort (n = 300) comprising adults aged 50–70 with established cardiovascular risk factors (hypertension, hyperlipidemia, diabetes mellitus, obesity, smoking history, or family history of premature coronary disease) but no prior diagnosis of coronary artery disease, heart failure, or atrial fibrillation; (2) an atrial fibrillation cohort (n = 100) with documented AF receiving standard anticoagulation therapy but without clinical stroke history; (3) a heart failure cohort (n = 100) with diagnosed heart failure (NYHA Class I–II, clinically stable) serving as a positive reference group for the cardiac-output-dependent linguistic signature; and (4) an age-matched control cohort (n = 150) with no more than one cardiovascular risk factor.

Baseline Assessment

All participants would undergo comprehensive baseline evaluation including: transthoracic echocardiography with ejection fraction quantification and diastolic function assessment; 12-lead electrocardiogram with assessment for AF, conduction abnormalities, and left ventricular hypertrophy; 24-hour Holter monitoring for paroxysmal arrhythmia detection and heart rate variability analysis; coronary artery calcium scoring by non-contrast CT; comprehensive metabolic and inflammatory panel (lipids, HbA1c, high-sensitivity CRP, IL-6, BNP/NT-proBNP); structural brain MRI with white matter hyperintensity quantification; quantitative EEG with spectral power analysis; and comprehensive neuropsychological testing battery with emphasis on processing speed, executive function, and attention.

Linguistic Data Collection

Participants would provide standardized narrative samples at monthly intervals for 48 months. Each session would include: a semi-structured autobiographical narrative (5-minute

CONFIDENTIAL — INVESTOR-2026

spoken account of a recent personally experienced event, audio-recorded in a standardized acoustic environment), a picture description task (Cookie Theft picture), a written narrative sample (typed dream report or autobiographical narrative), and daily 60-second voice diary entries via smartphone application for participants willing to provide high-frequency data. Spoken samples would include both acoustic-prosodic analysis and transcription-based linguistic analysis, as the prosodic channel is predicted to carry early indicators of autonomic dysregulation.

Follow-Up Assessment

Full cardiological, neuroimaging, and neuropsychological assessment would be repeated at 12, 24, 36, and 48 months. Heart rate variability analysis would be repeated at 6-month intervals to track autonomic trajectory. Interim cardiac events—including newly diagnosed AF, heart failure, acute coronary syndrome, coronary revascularization, new initiation of anticoagulation or heart failure therapy, and cardiac death—would be documented as primary endpoints. Interim cerebrovascular events would be documented as secondary endpoints to enable joint cardiac-cerebrovascular analysis.

Primary Analyses

The proposed analyses address five sequential questions.

First, cross-sectional validation of the cardiac linguistic signature: at baseline, do participants with established heart failure (Cohort 3) exhibit the predicted cardiac linguistic profile—executive-predominant cognitive-linguistic deficits with prosodic-autonomic features—distinguishable from age-matched controls (Cohort 4)? Do AF patients (Cohort 2) exhibit the predicted microembolic-burden linguistic profile?

CONFIDENTIAL — INVESTOR-2026

Second, prosodic-autonomic correlation: do baseline prosodic features (pitch variability, speech rhythm, respiratory patterning during speech) correlate with heart rate variability measures across all cohorts? This analysis tests the specific prediction that autonomic function is reflected in speech prosody.

Third, predictive detection: within the cardiac risk cohort (Cohort 1), do participants who subsequently experience cardiac events or newly diagnosed cardiac conditions during the 48-month follow-up show detectable linguistic trajectory changes before the clinical event? How far in advance do these changes become statistically significant?

Fourth, pathway discrimination: can the linguistic trajectory patterns associated with different cardiac pathologies (heart failure, AF, acute coronary events) be distinguished from each other and from the cerebrovascular trajectory pattern described in the companion paper (Beverly, 2026e)? This analysis tests the specificity claims of the five-pathway model.

Fifth, incremental predictive value: does linguistic trajectory data improve prediction of cardiac events beyond established risk factors (Framingham Risk Score, coronary calcium score, BNP levels)? This analysis addresses whether NOST-based monitoring adds clinically meaningful information to existing risk stratification.

Power Considerations

The cardiac risk cohort of 300 participants with monthly sampling over 48 months provides approximately 14,400 linguistic data points. Based on the expected cardiac event rate of 3–5% per year in a population with multiple risk factors (Greenland et al., 2010), the study would anticipate 36–60 primary cardiac endpoints over the follow-up period. The heart failure and AF reference cohorts provide approximately 4,800 additional data points each for cross-sectional signature validation. Mixed-effects modeling of individual trajectories, with linguistic

CONFIDENTIAL — INVESTOR-2026

features as time-varying predictors and cardiac events as time-to-event outcomes, would provide the primary statistical framework.

Clinical Implications

Functional Cardiovascular Screening

Current cardiovascular screening methods assess structural risk (coronary calcium, carotid intima-media thickness) or biochemical risk (lipids, inflammatory markers) but do not directly assess whether progressive cardiac pathology is producing functional end-organ consequences. NOST-based linguistic monitoring would provide a functional readout: not merely whether risk factors are present, but whether they are producing measurable cognitive impact. A patient with well-controlled hypertension whose linguistic trajectory remains stable provides different clinical information than a patient with the same blood pressure readings whose executive-linguistic profile is declining—the latter suggests that current management is insufficient to prevent end-organ consequences.

Atrial Fibrillation Detection and Monitoring

Paroxysmal atrial fibrillation is notoriously difficult to detect because episodes may be brief, asymptomatic, and absent during clinical encounters. The cumulative microembolic burden of undetected AF produces progressive cognitive damage that may precede rhythm diagnosis by years. If NOST-based monitoring can detect the cognitive-linguistic consequences of subclinical microembolic events, it could serve as an indirect AF detection pathway—identifying patients with unexplained progressive executive-linguistic decline who should undergo extended cardiac rhythm monitoring. This application is complementary to existing wearable rhythm monitoring

CONFIDENTIAL — INVESTOR-2026

(smartwatches, patch monitors) and could capture the cerebral consequences of arrhythmic episodes that occur between monitoring periods.

Heart Failure Early Detection

Heart failure with preserved ejection fraction (HFpEF) is increasingly recognized as a clinical challenge because standard echocardiographic screening may appear normal despite significant diastolic dysfunction and reduced cardiac output during exertion. The cognitive consequences of subclinical heart failure—documented even in asymptomatic compensated patients (Pressler et al., 2010)—represent a detectable functional marker. NOST-based monitoring could identify patients with declining linguistic trajectories suggestive of cardiac-output-dependent cognitive impairment, triggering referral for advanced cardiac assessment including exercise echocardiography and hemodynamic evaluation.

Integrated Cardiocerebrovascular Surveillance

The combination of this framework with the cerebrovascular detection framework (Beverly, 2026e) provides a comprehensive vascular surveillance system: linguistic monitoring that simultaneously screens for the cerebral consequences of both cardiac and cerebrovascular pathology. Because the predicted NOSS activation patterns differ between cardiac-origin and cerebrovascular-origin cognitive decline—particularly in the presence or absence of prosodic-autonomic features—the combined framework offers differential detection that existing screening methods cannot provide.

Limitations and Caveats

The limitations of this framework are substantial and must be acknowledged candidly.

CONFIDENTIAL — INVESTOR-2026

First, the framework is entirely theoretical. No empirical evidence currently exists demonstrating that linguistic monitoring detects cardiac pathology. The detection chain is logically coherent and each individual link is supported by published evidence, but the integrated system is untested.

Second, the cardiac-to-cerebral detection pathway involves more inferential steps than the direct cerebrovascular pathway described in the companion paper. Each additional step introduces potential noise and attenuation. The signal-to-noise ratio for detecting cardiac pathology through its cerebral linguistic consequences may prove insufficient for clinically useful detection—an empirical question that cannot be resolved theoretically.

Third, the multi-pathway model, while providing multiple potential detection channels, also creates a specificity challenge. Multiple cardiac conditions producing overlapping cerebral consequences through partially shared pathways may prove difficult to discriminate from each other and from non-cardiac causes of cognitive decline. The differential diagnostic value of the predicted pathway-specific signatures must be empirically validated before clinical claims are warranted.

Fourth, the prosodic-autonomic signature predicted for Pathway 3 represents the most novel and least empirically grounded prediction in this paper. While the relationship between HRV and cognitive function is well documented, and while autonomic innervation of laryngeal musculature is anatomically established, the specific prediction that cardiac autonomic dysregulation produces detectable prosodic changes in natural speech has not been directly tested. This prediction represents a genuinely novel hypothesis requiring dedicated empirical investigation.

CONFIDENTIAL — INVESTOR-2026

Fifth, the proposed 48-month study design is resource-intensive, and the expected cardiac event rate may prove insufficient for adequately powered predictive analyses. Collaborative multi-site designs, potentially integrated with existing cardiac risk cohort studies, would substantially improve feasibility and statistical power.

Sixth, the author's commercial interest in NOST-based applications constitutes a potential bias. Independent replication by cardiovascular research groups without commercial ties to the technology will be essential for establishing scientific validity. The proposed study would benefit significantly from collaboration with established cardiac research networks.

Conclusion

This paper proposes that progressive cardiovascular disease produces cerebral consequences that are detectable through NOST-based linguistic monitoring. Five pathophysiological pathways—reduced cardiac output, microembolic burden, autonomic dysregulation, systemic inflammation, and shared atherosclerotic burden—connect cardiac pathology to neural oscillatory changes that should manifest in the statistical properties of linguistic output. The predicted cardiac linguistic signature is composite, with prosodic-autonomic features preceding executive-cognitive features in the temporal sequence of disease progression, providing both detection and potential etiological characterization.

The clinical potential is significant. Cardiovascular disease kills more people worldwide than any other cause, and the majority of first cardiac events occur without prior clinical detection. If language-based monitoring can provide a functional early warning system—detecting the cerebral consequences of cardiac pathology months or years before acute events—it would address a critical gap in current cardiovascular screening and complement existing structural and biochemical risk assessment methods.

CONFIDENTIAL — INVESTOR-2026

The framework requires rigorous empirical validation before any clinical claims are warranted. The proposed prospective cohort study provides a structured research program for testing its predictions. The pathway-specific signatures, temporal sequencing predictions, and differential diagnostic claims are all specified with sufficient precision to be empirically confirmed or refuted. The scientific community's engagement—through collaboration, critique, and independent investigation—is essential to determining whether cardiocerebral coupling provides a viable detection pathway and whether the words we speak truly carry the signatures of the cardiac processes that sustain us.

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CONFIDENTIAL — INVESTOR-2026

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CONFIDENTIAL — INVESTOR-2026

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